

Chapter 14: Measurement and Effects of Heat

Textbook Exercise Solutions & Answers



1. Whom should I pair with?

Group 'A'	Group 'B'
1. Temperature of a healthy human body	b. 98.6 °F
2. Boiling point of water	d. 212 °F
3. Room temperature	a. 296 K
4. Freezing point of water	c. 0°C

2. Who is telling the truth?

(a) The temperature of a substance is measured in joule.

False. (The temperature of a substance is measured in °C or °F or K.)

(b) Heat flows from an object at higher temperature to an object at lower temperature.

True.

(c) The joule is the unit of heat.

True.

(d) Objects contract on heating.

False. (In general, objects expand on heating. There are some exceptions to this, you will learn about them in Standard X.)

(e) Atoms of a solid are free.

False. (Atoms of a solid are bound to each other due to the forces acting between them.)

(f) The average kinetic energy of atoms in a hot object is less than the average kinetic energy of atoms in a cold object.

False. (The average kinetic energy of atoms in a hot object is more than the average kinetic energy of atoms in a cold object.)

3. You will find if you search.

(a) A thermometer is used to measure

A thermometer is used to measure temperature.

(b) The apparatus used to measure heat is called a

The apparatus used to measure heat is called a calorimeter.

(c) Temperature is the measure of the kinetic energy of the atoms in a substance.

Temperature is the measure of the average kinetic energy of the atoms in a substance.

(d) The heat contained in a substance is the measure of the kinetic energy of the atoms in the substance.

The heat contained in a substance is the measure of the total kinetic energy of the atoms in the substance.

4. Nishigandha kept a vessel containing all the ingredients for making tea in a solar cooker. Shivani kept a similar vessel on a stove. Whose tea will be ready first and why?

Shivani's tea will be ready first.

Reason: In a given time, the amount of heat received by the vessel on a stove is far greater than that received by the vessel kept in a solar cooker.

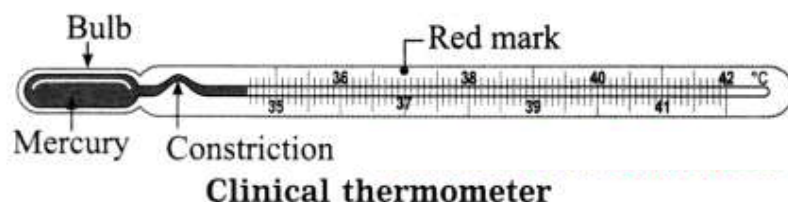
5. Write brief answers.

(a) Describe a clinical thermometer. How does it differ from the thermometer used in a laboratory?

A clinical thermometer has a narrow stem and a long bulb filled with mercury (or alcohol). There is a small constriction in the stem above the bulb. When the bulb of the thermometer is held in the armpit or the mouth of a patient, the mercury (or alcohol) in the bulb rises in the stem.

When it is taken out of the patient's body, the small constriction does not allow the mercury (or alcohol) from the stem to retreat into the bulb. Thus, this arrangement enables us to read the temperature of the patient's body at ease after the removal of the thermometer from his body.

The clinical thermometer is graduated from 35°C to 42°C (or from 95°F to 108°F). At 37°C (98.6°F), there is a red arrow mark which indicates the temperature of a healthy person.



The thermometer used in a laboratory has wider range and does not have constriction like a clinical thermometer.

(b) What is the difference between heat and temperature?

Heat is related to the total kinetic energy of the atoms in a substance while temperature is related to the average kinetic energy of the atoms in the substance. Heat flows from a body at higher temperature to a body at lower temperature. Thus, temperature is a quantity that determines the direction of flow of heat. It is a quantitative measure of the degree of hotness or coldness of a body.

Higher temperature does not mean higher heat content. Suppose a vessel A contains 2 litres of water at 90°C and a vessel B contains 100 ml of water at 91°C . Then the heat content of water in A is greater than that of water in B, but the temperature of water in B is higher than that of water in A.

Units of heat:

Heat is usually expressed in calorie, kilocalorie and joule. It can also be expressed in erg as heat is a form of energy.

(c) Explain the construction of a calorimeter. Draw the necessary figure.

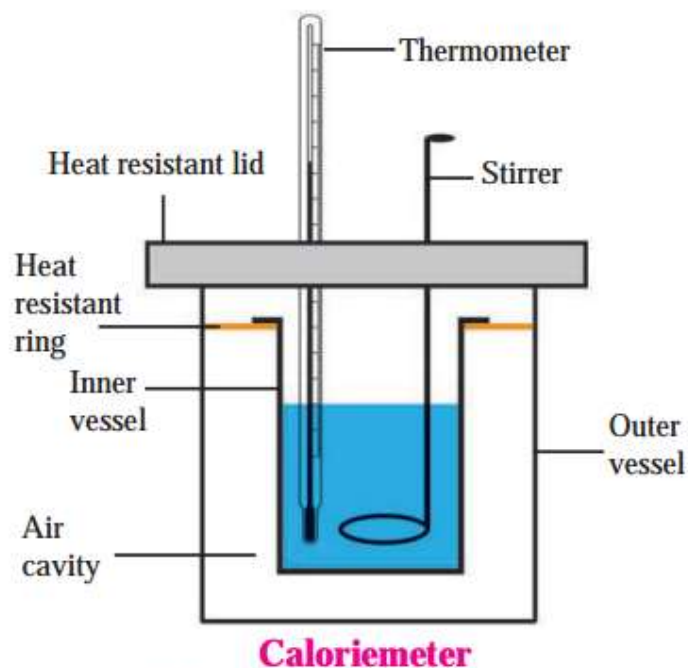


Figure shows the construction of a calorimeter. Like a thermo flask, a calorimeter has two vessels. The inner vessel, made of copper, is (practically) thermally isolated from the surroundings. The outer vessel is made of wood and is covered with a heat resistant lid. The lid has two holes, one for the thermometer and the other for the stirrer. The inner and outer surfaces of the inner vessel are polished for minimizing exchange of heat with the surroundings by radiation. A heat resistant ring covers the inner vessel.

(d) Explain why rails have gaps at specific distances.

The rails expand in summer due to increase in the temperature of the atmosphere. Also, they expand due to rise in temperature caused by the friction between the rails and the wheels of the running train. If there is no gap between successive rails of a railway line, the rails would bend due to expansion. This bending and twisting of the rails would cause accidents. Hence, a gap is kept between successive rails of a railway line to make provision for their expansion in summer.

(e) Explain with the help of formulae the expansion coefficients of liquid and gas.

1. A liquid is held in a container. When it is heated, both the container and the liquid expand. The expansion of the container is usually very small compared to that of the liquid in it. Often, it can be ignored.

Suppose a liquid is heated so that its temperature rises by ΔT (very small) and its volume increases from V_1 to V_2 . Experimentally, it is found that the increase in volume, $V_2 - V_1$, is proportional to V_1 and ΔT .

Hence, $(V_2 - V_1) \propto V_1 \Delta T$.

$\therefore V_2 - V_1 = \beta V_1 \Delta T$, where β is a constant of proportionality called the volumetric expansion coefficient of the liquid.

$$\beta = \frac{V_2 - V_1}{V_1 \Delta T}.$$

It is expressed in per $^{\circ}\text{C}$.

We have $V_2 = V_1 + \beta V_1 \Delta T = V_1(1 + \beta \Delta T)$.

β is the increase in the volume of a liquid per unit original volume per unit rise in its temperature.

2. A gas is enclosed in a container. When it is heated at constant pressure, both the container and the gas expand. Suppose a gas is heated at constant pressure so that its temperature rises by ΔT (very small) and its volume increases from V_1 to V_2 . Experimentally, it is found that the increase in volume, $V_2 - V_1$ is proportional to V_1 and ΔT . Hence, $(V_2 - V_1) \propto V_1 \Delta T$.

$\therefore V_2 - V_1 = \beta V_1 \Delta T$, where β is a constant of proportionality, the volumetric expansion coefficient, called the constant pressure expansion coefficient.

$$\beta = \frac{V_2 - V_1}{V_1 \Delta T}.$$

It is expressed in per $^{\circ}\text{C}$.

We have $V_2 = V_1 + \beta V_1 \Delta T = V_1(1 + \beta \Delta T)$.

β is the increase in the volume of a gas per unit original volume per unit rise in its temperature when the pressure is kept constant.

6. Solve the following examples:

(a) What must be the temperature in Fahrenheit so that it will be twice its value in Celsius?

Given:

$$F = 2C \implies C = \frac{F}{2}$$

Formula:

$$\frac{F - 32}{9} = \frac{C}{5}$$

Calculation:

Substitute $C = \frac{F}{2}$ into the formula:

$$\frac{F - 32}{9} = \frac{F/2}{5} = \frac{F}{10}$$

Cross-multiplying:

$$10(F - 32) = 9F$$

$$10F - 320 = 9F$$

$$10F - 9F = 320 \implies F = \mathbf{320^\circ F}$$

Answer:

The temperature in Fahrenheit must be **320 °F** (which equals 160 °C).

(b) A bridge is made from 20 m long iron rods. At temperature 18 °C, the distance between two rods is 0.4 cm. Up to what temperature will the bridge be in good shape?

Given:

Initial length of iron rod (l_1) = 20 m

Permissible expansion ($\Delta l = l_2 - l_1$) = 0.4 cm = 4×10^{-3} m

Initial temperature (T_1) = 18°C

Coefficient of linear expansion of iron (λ) = $11.5 \times 10^{-6}/^\circ\text{C}$

Formula:

$$\Delta l = \lambda \cdot l_1 \cdot \Delta T \implies \Delta T = \frac{\Delta l}{\lambda \cdot l_1}$$

Calculation:

$$\Delta T = \frac{4 \times 10^{-3} \text{ m}}{(11.5 \times 10^{-6}/^\circ\text{C}) \times 20 \text{ m}} = \frac{4 \times 10^{-3}}{230 \times 10^{-6}} = \frac{400}{23} ^\circ\text{C} \approx 17.39^\circ\text{C}$$

Now, final temperature (T_2):

$$T_2 = T_1 + \Delta T = 18^\circ\text{C} + 17.39^\circ\text{C} = \mathbf{35.39^\circ\text{C}}$$

Answer:

The bridge will be in good shape up to a temperature of **35.39 °C**.

(c) At 15 °C the height of Eiffel Tower is 324 m. If it is made of iron, what will be the increase in length in cm, at 30 °C?

Given:

Initial height of Eiffel Tower (l_1) = 324 m

Initial temperature (T_1) = 15°C

Final temperature (T_2) = 30°C

Rise in temperature ($\Delta T = T_2 - T_1$) = $30^\circ\text{C} - 15^\circ\text{C} = 15^\circ\text{C}$

Coefficient of linear expansion of iron (λ) = $11.5 \times 10^{-6}/^\circ\text{C}$

Formula:

$$\Delta l = \lambda \cdot l_1 \cdot \Delta T$$

Calculation:

$$\Delta l = (11.5 \times 10^{-6}/^\circ\text{C}) \times 324 \text{ m} \times 15^\circ\text{C} = 55,890 \times 10^{-6} \text{ m} = 0.05589 \text{ m}$$

Converting to centimeters (1 m = 100 cm):

$$\Delta l = 0.05589 \times 100 \text{ cm} = \mathbf{5.589 \text{ cm} \approx 5.6 \text{ cm}}$$

Answer:

The increase in the height of the Eiffel Tower at 30 °C is **5.589 cm** (approx. **5.6 cm**).

(d) Two substances A and B have specific heats c and $2c$ respectively. If A and B are given Q and $4Q$ amounts of heat respectively, the change in their temperatures is the same. If the mass of A is m , what is the mass of B?

Given:

Substance A: Specific heat $c_A = c$, Heat supplied $Q_A = Q$, Mass $m_A = m$

Substance B: Specific heat $c_B = 2c$, Heat supplied $Q_B = 4Q$, Mass $m_B = ?$

Temperature change: $\Delta T_A = \Delta T_B = \Delta T$

Formula:

$$Q = m \cdot c \cdot \Delta T \implies \Delta T = \frac{Q}{m \cdot c}$$

Calculation:

For substance A:

$$\Delta T = \frac{Q}{m \cdot c} \quad \text{--- (Equation 1)}$$

For substance B:

$$\Delta T = \frac{4Q}{m_B \cdot (2c)} = \frac{2Q}{m_B \cdot c} \quad \text{--- (Equation 2)}$$

Equating (1) and (2) since ΔT is the same:

$$\frac{Q}{m \cdot c} = \frac{2Q}{m_B \cdot c}$$

Dividing both sides by $\frac{Q}{c}$:

$$\frac{1}{m} = \frac{2}{m_B} \implies m_B = \mathbf{2m}$$

Answer:

The mass of substance B is **2m** (twice the mass of substance A).

(e) When a substance having mass 3 kg receives 600 cal of heat, its temperature increases by 10 °C.

What is the specific heat of the substance?

Given:

Mass of substance (m) = 3 kg = 3,000 g

Heat received (Q) = 600 cal

Temperature rise (ΔT) = 10°C

Formula:

$$Q = m \cdot c \cdot \Delta T \implies c = \frac{Q}{m \cdot \Delta T}$$

Calculation:

In CGS units:

$$c = \frac{600 \text{ cal}}{3,000 \text{ g} \times 10^\circ\text{C}} = \frac{600}{30,000} = \mathbf{0.02 \text{ cal/(g} \cdot ^\circ\text{C)}}$$

In SI units (1 cal = 4.184 J):

$$c = \frac{600 \times 4.184 \text{ J}}{3 \text{ kg} \times 10 \text{ K}} = \mathbf{83.68 \text{ J/(kg} \cdot \text{K)}}$$

Answer:

The specific heat of the substance is **0.02 cal/(g·°C)** (or **83.68 J/(kg·K)**).